Coordinated Use of Smart Inverters With Legacy Voltage Regulating Devices in Distribution Systems With High Distributed PV Penetration—Increase CVR Energy Savings

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***Abstract*—This paper studies the coordinated use of smart inverters with legacy voltage regulating devices to help increase the energy savings from conservation voltage reduction (CVR) in distribution systems with high photovoltaic penetration. Two voltage optimization algorithms are developed to evaluate the CVR benefit by considering two types of smart inverter functions: aggregated reactive power control and autonomous volt/VAR control. In each algorithm, smart inverters are coordinately controlled with the operation of legacy voltage regulating devices including load tap changers and capacitor banks. Both algorithms are tested on representative utility distribution system models. Simulation results demonstrate that the proposed algorithms can achieve around 1.8%–3.6% energy savings when only using legacy voltage regulating devices and an additional 0.3%–0.9% when adding smart inverters.**

***Index Terms*—Photovoltaic, smart inverter, volt/VAR, CVR, reactive power control, energy saving, power quality.**

I. Introduction

DECLINING system costs and various incentive mechanisms have stimulated the installation of distributed energy resources (DERs). Traditionally, distributed solar photovoltaics (PV) are installed with standard inverters that only output real power. Recently, however, PV is increasingly being paired with smart inverters that can both supply and absorb reactive power, and thus distributed PV has the potential to support and influence local voltage and power factor on the grid in the form of volt/VAR control (VVC). Smart inverter VVC can be achieved using either autonomous mode or aggregated mode. In autonomous mode, smart inverter decides its reactive power output based on its local voltage by completely following the predefined volt/VAR curve [1]–[3]. In aggregated mode, the operational power factor or the reactive power output of the smart inverter is determined by the control signal sent from the control algorithm [4]–[7].

Conservation voltage reduction (CVR) is a method of lowering voltage on a circuit to conserve energy and reduce peak load. Utilities can control their distribution system voltage profiles by controlling the legacy voltage regulating devices, such as load tap changers (LTCs), voltage regulators, and shunt capacitors within the substation and distribution system. ANSI Standard C84.1-2011 [8] stipulates that during normal operations the voltage at the customer premise must be limited between 0.95 and 1.05 p.u., referred to as *ANSI limit* in this paper. Traditionally, without PV distribution system voltage decreases as the distance from the upstream voltage regulating equipment increases, so the substation voltage is maintained closer to 1.05 p.u. to guarantee that customers at the end of the feeder have service voltage within the ANSI limit; however, it has been shown that a 1% reduction in distribution service voltage can drive a load reduction from 0.3% to 1% [9]. Early CVR tests date to the 1970s [10]. Many utilities—including Southern California Edison [11], Northeast Utilities [12], Bonneville Power Administration [13], and BC Hydro [14]—have conducted various tests and demonstrated the effectiveness of CVR on saving energy consumption and reducing peak demand. Some recent CVR studies were also conducted because of the increasing interest in energy savings. The study conducted by the Electric Power Research Institute in [15] quantified losses on the distribution system and found that voltage reduction "produces the majority of savings" in energy consumption. The technical report from the Pacific Northwest National Laboratory found that typical savings from a CVR scheme range from 0.5% to 4% of energy consumption and peak reduction on individual distribution circuits [16].

Most existing CVR studies rely on controlling legacy voltage regulating devices but do not include the impact of DERs because DERs have not previously been allowed to participate in voltage regulation; however, in 2014 an amendment was made to the major DER interconnection standard (IEEE 1457a [17]) to allow inverter-based generation to

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participate in distribution feeder voltage regulation. Different from traditionally standard inverters that can only operate at unity power factor, smart inverters can control both active power and reactive power outputs using different smart inverter functions, such as volt/VAR control, volt/watt control, power factor control, etc. With the development of PV inverters and the requirements of PV interconnection standards, smart inverter has been widely adopted by PV system vendors and most existing and future PV inverters are expected to support smart inverter functions. By adding distributed PV with smart inverters at various locations on a distribution circuit, an opportunity exists to control the secondary voltages and, in aggregate, the primary voltage in a way not previously possible using traditional CVR schemes controlling centralized voltage control equipment.

Thus, this paper studies the new types of CVR schemes by coordinately controlling legacy voltage regulating devices and smart inverters, and evaluates the capability of smart inverters coordinating with legacy voltage regulating devices on increasing energy savings. In order to model and analyze the impact of both autonomous and aggregated inverter functions on CVR energy saving, two voltage optimization methodologies are developed to complement the traditional CVR scheme by respectively incorporating the autonomous volt/VAR control and aggregated reactive power control. Less communications are needed for the one with autonomous volt/VAR control because all smart inverters are acting autonomously based on their local voltages. The proposed methodologies are tested on two utility distribution system models. The distinctions in utilities, voltage levels, and peak loads provide a range of characteristics to analyze the impact of distributed PV with smart inverters.

The paper is organized as follows. Section II discusses the load model used in the study, and Section III presents the two CVR voltage optimization methodologies. Section IV provides the simulation studies and result discussions. Finally, conclusions are given in Section V.

II. Load Modeling

An exact assessment of energy savings requires accurate load modeling. Loads can be modeled into two distinct classes: those without thermal cycles (using the ZIP model) and those with thermal cycles, such as heating, ventilating, and air-conditioning systems (using the equivalent thermal parameter [ETP] model) [16]; and both ZIP and ETP models reflect the voltage-dependent characteristics. Detailed load information is generally unknown for most feeders, and thus it is hard to explicitly model loads using the ZIP and ETP models without enough observability. An alternative approach to model loads by accommodating the complex end-use behaviors in a practical manner specifies how active and reactive load power vary with voltage using the CVR factor [15].

The CVR factor can be defined as a reduction of power based on a voltage variation, given as [18]:

$$CVR_f = \frac{\%reduction\ in\ kW/kVAR}{\%reduction\ in\ voltage} \tag{1}$$

Same as the load model used in [15], this paper uses the following exponential equation to model the relationship between load power and voltage, and the power CVR factor is used as the exponential parameter.

$$\begin{cases} P^i_{load}(t) = P^i_n \cdot V_i(t)^{CVR_{f(kW)}} \\ Q^i_{load}(t) = Q^i_n \cdot V_i(t)^{CVR_{f(kvar)}} \end{cases} \tag{2}$$

where $CVR_{f(kW)}$ is the active power CVR factor and $CVR_{f(kvar)}$ is the reactive power CVR factor. $P^i_{load}(t)$ and $Q^i_{load}(t)$ are the active power and reactive power, respectively, of the load connecting at bus i at time point t. P^i_n and Q^i_n are the active power and reactive power, respectively, of the load connecting at bus i under the nominal voltage. $V_i(t)$ is the voltage magnitude of bus i at time point t.

The CVR factors of different distribution feeders are not unique. Some typical CVR factor values are summarized in [18], showing that $CVR_{f(kW)}$ can range from 0.16 to 1.19, and $CVR_{f(kVar)}$ can range from 1.99 to 20.12.

III. CVR Voltage Optimization Methodology

A. Measuring the CVR Effect

In this paper, the CVR voltage optimization effect is measured using the energy saving, which is the total load energy consumption reductions, defined as:

$$CVR\ saving(\%) = \frac{Base\ Energy - Energy_{new}}{Base\ Energy} \cdot 100\% \tag{3}$$

where $CVRsaving$ is the percentage energy savings achieved, $BaseEnergy$ is the total load energy consumptions for the base case when there is no PV or voltage optimization, and $Energy_{new}$ is the total load energy consumption after implementing voltage optimization.

B. Voltage Reduction

According to (2), load demand is proportional to the voltage magnitude at each time point. Thus, voltage reduction is implemented to obtain the minimum energy consumption, and this problem can be formulated as:

$$\min J = Energy_{CVR}(t) = \sum_{i=1}^{NL} P^i_{load}(t) \tag{4}$$

where $Energy_{CVR}(t)$ is the total energy consumption at time point t. NL is the total number of loads.

When implementing the voltage optimization, all bus voltages must be within the ANSI limit and thermal constraint should be satisfied, as:

$$0.95 \le V_i(t) \le 1.05, \quad i = 1, 2, \ldots, N \tag{5}$$

$$I_k(t) \le I_{lim,k}, \quad k = 1, 2, \ldots, M \tag{6}$$

where $V_i(t)$ represents the per-unit voltage of bus i at time point t. N is the total number of nodes. $I_k(t)$ is the current in branch k at time t, and $I_{lim,k}$ is the maximum current limit for branch k. M is the total number of branches.

In addition, power flow equations should always be satisfied for each time step, as:

$$\begin{cases} P_i^p = \sum_{j=1}^{N}\sum_{l=a}^{c}\left[e_i^p\left(g_{ij}^{pl}\cdot e_j^l - b_{ij}^{pl}\cdot f_j^l\right) + f_i^p\left(g_{ij}^{pl}\cdot f_j^l + b_{ij}^{pl}\cdot e_j^l\right)\right] \\ Q_i^p = \sum_{j=1}^{N}\sum_{l=a}^{c}\left[f_i^p\left(g_{ij}^{pl}\cdot e_j^l - b_{ij}^{pl}\cdot f_j^l\right) - e_i^p\left(g_{ij}^{pl}\cdot f_j^l + b_{ij}^{pl}\cdot e_j^l\right)\right] \end{cases} \tag{7}$$

where P_i^p and Q_i^p are the active power injection and reactive power injection at bus i respectively. p denotes the phase, which can be phase a, b, or c. $y_{bus,ij}^{pl} = g_{ij}^{pl} + j\cdot b_{ij}^{pl}$ is the matrix element of the bus admittance matrix $\mathbf{Y_{bus}}$. $V_i^p = e_i^p + j\cdot f_i^p$ is the phase-p voltage at bus i.

Traditionally, controlling and lowering distribution voltage is achieved by optimizing capacitor operation and lowering the output voltage of LTCs and voltage regulators. The operation of these legacy voltage regulating devices can be formulated as:

$$\begin{cases} cap_i(t) = 0 \text{ or } 1, & i = 1, 2, \ldots, NC \\ -16 \le tap_j(t) \le 16, & j = 1, 2, \ldots, NR \end{cases} \tag{8}$$

where $cap_i(t)$ represents the state of the *i-th* capacitor at time point *t:* 0 means the capacitor is switched off, and 1 means it is switched on. $tap_j(t)$ represents the *j-th* regulator tap position at time t, and its maximum and minimum values are 16 and −16 for the typical use of the 32-step regulator. *NC* and *NR* are the total number of capacitors and regulators, respectively.

Smart PV inverters can control their output power to affect the feeder voltage. This paper assumes that the reactive power capability of the smart inverter follows current industry best practices, which is limiting the amount of VAR to 60% of the inverter capacity for VVC. This means that the minimum power factor limit is 0.8 when the inverter operates at its kVA rating. In addition, two methods produce reactive power for smart inverters: watt priority and VAR priority. Under VAR-priority mode, the production of reactive power is prioritized over the production of real power, and real power will be curtailed if there is not sufficient headroom in the inverter. This paper studies the VAR-priority mode for smart inverters. Thus, when PV output is less than 80% of the nameplate kVA rating, no curtailment is needed; but when PV output is greater than 80%, active power curtailment might be needed to supply reactive power. Fig. 1 illustrates the operational range of the reactive power for a smart inverter. This paper assumes that the active power curtailment is limited to 10% of the inverter kVA rating. Note that if a larger curtailment threshold is used, more energy savings are expected.

As a result, the operational constraints of smart inverters are formulated as:

$$\begin{cases} Q_{SI,i}(t) = \alpha_{SI,i}(t)\cdot S_{SI,i} & \text{and } 0 \le \left|\alpha_{SI,i}(t)\right| \le 0.6 \\ P_{SI,i}^{AC}(t) = P_{SI,i}^{DC}(t), & \text{if } P_{SI,i}^{DC}(t) \le 0.8\cdot S_{SI,i} \\ P_{SI,i}^{AC}(t) = P_{SI,i}^{DC}(t) - P_{SI,i}^{curt}(t), & \text{if } P_{SI,i}^{DC}(t) > 0.8\cdot S_{SI,i} \\ P_{SI,i}^{curt}(t) \le 0.1\cdot S_{SI,i} & \\ i = 1, 2, \ldots, NS. & \end{cases} \tag{9}$$

where $Q_{SI,i}(t)$ is the reactive power output of smart inverter i at time point t, and it can be both positive (generating reactive

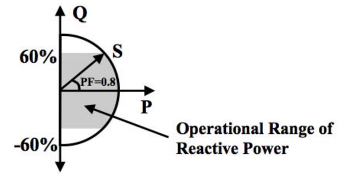

Fig. 1. Operational range of the reactive power for a smart PV inverter.

power) and negative (consuming reactive power). $S_{SI,i}$ is the nameplate kVA rating of smart inverter i. $\alpha_{SI,i}(t)$ is the reactive power coefficient of smart inverter i at time point t. $P_{SI,i}^{DC}(t)$ and $P_{SI,i}^{AC}(t)$ are the maximum power output of the PV system i and the actual AC active power output from the PV inverter at time point t, respectively. $P_{SI,i}^{curt}(t)$ is the active power curtailment for the PV inverter i at time point t. *NS* is the total number of smart PV inverters.

C. Voltage Optimization Algorithms

Although the essence of smart inverter is controlling its active and reactive power outputs, the way of altering output power depends on the smart inverter function that is in use. In order to evaluate the impact of two types of smart inverter functions on CVR benefit, two voltage optimization algorithms are proposed to coordinately control LTCs, shunt capacitors, and smart inverters to lower the voltage profile and achieve maximum CVR energy savings. In aggregated reactive power control, the reactive power coefficients of smart inverters, defined in (9), are determined together with the operational status of legacy voltage regulating devices by simultaneously solving an integrated optimization problem. In the autonomous VVC, the reactive power coefficients are determined from the predefined volt/VAR curve based on the local voltages, and the optimal operations of legacy voltage regulating devices are solved in coordination with the autonomous VVC effect. Fig. 2 shows the integrated flowchart of the two voltage optimization algorithms. To differentiate the energy saving effects of controlling the legacy voltage regulating devices and using smart inverters, the proposed algorithms are divided into two stages: legacy voltage control and smart inverter participation. The left side of Fig. 2 shows the smart inverter participation approach used in Algorithm 1, and the right side shows the one used in Algorithm 2. Because energy consumption is used to measure the CVR effect, the quasi-static time-series (QSTS) study is considered in both algorithms. Each algorithm is implemented every time step based on the load and PV data measured at that time. Details about both algorithms are provided below.

1) Algorithm 1 (Use Autonomous Volt/VAR Control): Smart inverters participate in Algorithm 1 using their autonomous VVC and coordinate with the operations of LTC and shunt capacitors. The entire algorithm is based on three parts, including capacitor optimization, LTC optimization, and autonomous VVC.

a) Legacy voltage control: As shown in Fig. 2, initially all smart inverter controls are disabled, and the legacy voltage control is activated by implementing the capacitor optimization

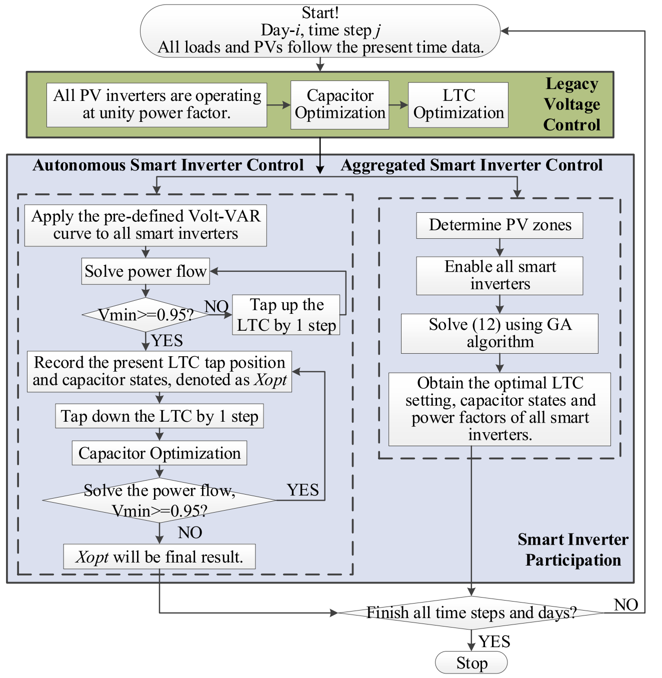

Fig. 2. Flowchart of the proposed CVR voltage optimization algorithms using autonomous and aggregated smart inverter controls.

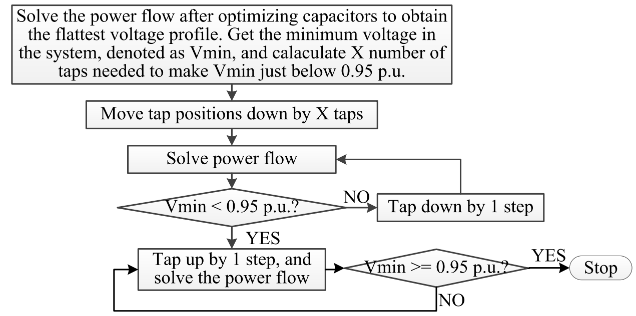

Fig. 3. Heuristic algorithm for LTC optimization.

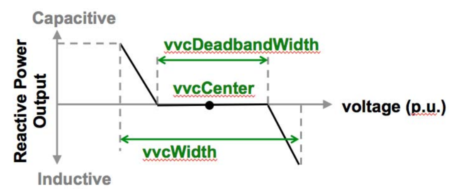

Fig. 4. Typical volt/VAR curve used for autonomous VVC.

to flatten the voltage and the LTC optimization to lower the voltage.

The depth of the voltage reduction is limited for feeders that experience significant voltage drop, so a relatively flat voltage profile along the feeder is preferable to achieve more voltage reduction and more CVR benefit [9]. Thus, capacitor optimization is first conducted to achieve the flattest voltage profile, i.e., the smallest difference between the maximum and minimum voltages in the feeder. All possible capacitor states (switched ON or OFF per capacitor) are studied, and the state that leads to the smallest voltage difference in the system will be considered the optimal solution.

Then the LTC tap position is optimized by selecting the lowest possible position without causing a voltage violation less than 0.95 p.u. A heuristic algorithm as shown in Fig. 3 is proposed to implement the LTC optimization.

b) Autonomous smart inverter volt/VAR control: The autonomous VVC regulates the reactive power output of the smart inverter based on its local voltage by following a volt/VAR curve. Fig. 4 shows a typical volt/VAR curve, which is determined by three parameters, including the voltage center (vvcCenter), which is the desired voltage; the voltage width (vvcWidth), determining the slope of the volt/VAR curve and how much reactive power the inverter provides to move the terminal voltage toward vvcCenter; and the deadband (vvcDeadbandWidth), defining the voltage range around vvcCenter in which the inverter does not provide any reactive power. In general, the values of these three volt/VAR curve parameters are provided by smart inverter vendors. The values used in this paper are selected from a group of candidates to achieve the largest energy savings, and the selection approach can refer to [19].

After implementing the legacy voltage control, the volt/VAR curve, determined in advance, is applied to all smart inverters to activate the autonomous VVC, and the power flow is solved. If the lowest voltage is less than 0.95 p.u., the LTC is tapped up until the lowest voltage exceeds 0.95 p.u. Then the iterative operations of the LTC and capacitors are performed to seek a possible additional voltage reduction: (a) implementing the capacitor optimization to flatten the voltage again; (b) tapping down the LTC by one step. Such iterative operations will stop if the minimum voltage is less than 0.95 p.u. with any additional LTC tapping down.

2) Algorithm 2 (Use Aggregated Reactive Power Control): Similar to Algorithm 1, Algorithm 2 first uses the legacy voltage regulating devices to flatten and lower the voltage profile. Then the reactive power outputs of all smart inverters are determined by solving an optimization problem.

Separately controlling individual smart inverters will cause more communication requirements, so this paper studies smart inverters to be controlled in an aggregated manner. Multiple PV zones are first determined; the smart inverters that are within the same zone are controlled aggregately and thus assumed to have the same reactive power coefficients, but different zones have different coefficients. PV zones are divided according to the voltages at PV buses. When using smart inverters to lower the voltage, it is expected that the smart inverter can generate less or consume more reactive power. Thus, if the PV bus voltage is already low, less reactive power consumption is needed; but if the PV node voltage is high, less reactive power output or more reactive power consumption is expected to achieve more voltage reductions. In sum, PV zones are divided by following the procedure below.

(a) Solve the power flow and get the voltages at all PV buses, and find the maximum value and minimum value, denoted as $MaxV_{PV}$ and $MinV_{PV}$, respectively.

(b) Specify the total number of PV zones that we would like to have, denoted as *NZ*. It is noted that *NZ* is a pre-defined value, which can range from 1 to the total number of PV systems. The voltage interval is computed:

$$\Delta Vrange = \frac{MaxV_{PV} - MinV_{PV}}{NZ} \tag{10}$$

Then the total *NZ* voltage ranges are defined as:

$$\begin{aligned} &[MinV_{PV}, MinV_{PV} + \Delta Vrange) \\ &\quad [MinV_{PV} + \Delta Vrange, MinV_{PV} + 2 \cdot \Delta Vrange) \\ &\qquad \vdots \\ &\quad \left[MinV_{PV} + (NZ - 1) \cdot \Delta Vrange, MinV_{PV} \right. \\ &\qquad \left. + NZ \cdot \Delta Vrange\right] \end{aligned} \tag{11}$$

(c) Find all PV systems with the voltages within each of the total *NZ* ranges, and those PV systems will constitute a PV zone.

Note that because PV bus voltages are time varying, the PV zones are dynamically identified for every time step during the QSTS simulation. That is to say, at every time step, PV bus voltages are checked after solving power flow, and all PV systems are divided into *NZ* groups as (11).

Then the smart inverter reactive power output, capacitor states, and LTC tap position will be determined by solving the following mixed-integer nonlinear optimization problem:

$$\begin{aligned} \min \;\; & J(u_t) = Energy_{CVR}(t) \\ s.t. \;\; & (2), (5) - (9) \end{aligned} \tag{12}$$

where u_t represents the decision variables, including zonal smart inverter reactive power coefficients, capacitor states, and LTC tap position at the present time, t.

This paper proposes a genetic algorithm to solve the above optimization problem. The genetic algorithm is a meta-heuristic inspired by the process of natural selection; and it can generate high-quality solutions to optimization by relying on bio-inspired operators, including mutation, crossover, and selection. A population of candidate solutions is evolved toward better solutions. Although the genetic algorithm cannot guarantee global optimality, it is effective to solve the mixed integer nonlinear optimization problem and provide the near-optimal solution [20], [21]. Fig. 5 shows the flowchart of the proposed genetic algorithm, which is coordinating MATLAB codes and OpenDSS [22] power flow solver to solve (12).

Initialization will generate the first-generation population with all chromosomes to be feasible, i.e., satisfying all constraints. OpenDSS is used to model the distribution system and PV systems. MATLAB codes are developed to modify the OpenDSS model automatically based on the generated chromosome, including capacitor states, LTC tap position, and reactive power coefficients of all smart inverters.

Crossover operator randomly selects two chromosomes and swaps their information to create two new ones. To accommodate the hybrid genes mixed of integers and continuous values, an approach to conduct crossover has been proposed by Ding and Loparo [21] based on the combination of single-point crossover and arithmetic crossover, and it can be used

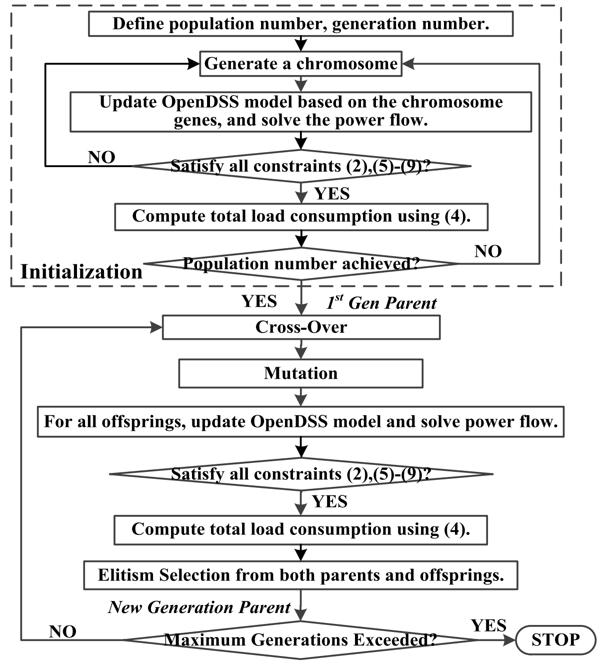

Fig. 5. Flowchart of the proposed genetic algorithm approach.

in this paper for the proposed study. The mutation operator randomly changes one gene in the selected chromosome into another feasible value to introduce new information into the offspring.

Then the fitness values of all parents and offspring are computed using (4), and elitism is used to select the best population. The entire procedure is repeated until the maximum generation exceeded.

D. Practical Application

The proposed CVR study is a planning study to understand the impact of coordinated operations between legacy voltage regulating devices and smart PV inverters on increasing CVR energy savings. The data inputs required to achieve the proposed study include historical AMI measurements of loads and PVs, SCADA measurements at feeder head and distribution feeder model, which can be acquired from utilities. The proposed CVR study is generic and applicable to all distribution feeders.

In addition, the proposed CVR voltage optimization approaches can be used for distribution grid operations. With the aid of real-time communications and forecasted loads and PVs, the proposed two voltage optimization algorithms can be implemented to solve the optimal setpoints for both legacy voltage regulation devices and smart inverters for the next control time window (such as 15 mins). The proposed voltage optimization algorithms can also interface with existing advanced distribution management systems (ADMS) to control field devices.

E. CVR Voltage Optimization Approach Discussion

Algorithm-1, based on autonomous VVC, doesn't require communications between legacy voltage regulating devices

TABLE I
TEST DISTRIBUTION SYSTEM INFORMATION

Model Property	System 1	System 2
Substation Bank Size	10 MVA	45 MVA
Circuit Primary Voltage	12 kV	21 kV
Number of Feeders	2	3
Number of primary buses	1,124	2,197
Peak Load	6.3 MW	37.1 MW
Max. Circuit Distance	9 miles	6.63 miles
Capacitor Banks	0	7
LTC Setting	Base=120 V, R=7, X=7, CTPrim=600	Base=122 V, R=4, X=0, CTPrim=1,500

and smart inverters, or communications among inverters. All smart inverters determine their reactive power output based on local voltage measurements autonomously. The statuses of LTC and capacitor banks are solved at each time snapshot during QSTS study based on the power flow results, so the operations of fast-acting inverters and slow-acting legacy voltage regulating devices are indeed decoupled. On the other hand, Algorithm-2, based on aggregated reactive power control, is updating smart inverter reactive power coefficients, LTC tap position and capacitor states simultaneously, and it is implemented in a centralized manner in this paper so the communication is required between aggregated decision-making and individual devices. The implementation of distributed control could help solve the communication issue. Also, although smart inverters can be controlled in a faster time granularity, this paper doesn't consider the time differentiation between smart inverter operations and legacy voltage regulating device operations. To capture different operational time scales, the hierarchical control can be implemented so that smart inverter reactive power coefficients can be updated in a faster pace. Both communication and time differentiation will be addressed in our future study.

IV. CASE STUDY

The proposed two CVR voltage optimization approaches are tested on two representative utility distribution systems. Table I lists key characteristics of the two systems. These two systems are modeled starting from the substation transformer at the primary circuit down to the customers at the secondary circuit. In addition, one LTC on the secondary side of the transformer is modeled with line-drop compensation for each system, and it regulates the voltage for all feeders included in the system. The topologies of both systems are provided in Fig. 6.

The CVR factor of each distribution feeder is different, and detailed field experiments are needed to obtain the actual values. In [15], a CVR factor of 0.8 was used for watts and 3.0 was used for Vars, and these two values are used in this paper for the following simulation study. If the CVR factor for the load type studied is known to be higher or lower, an increase or decrease in energy savings is expected.

A. PV System Modeling

In this paper, distributed PV systems are added to the distribution system models. PV systems are sized based on an

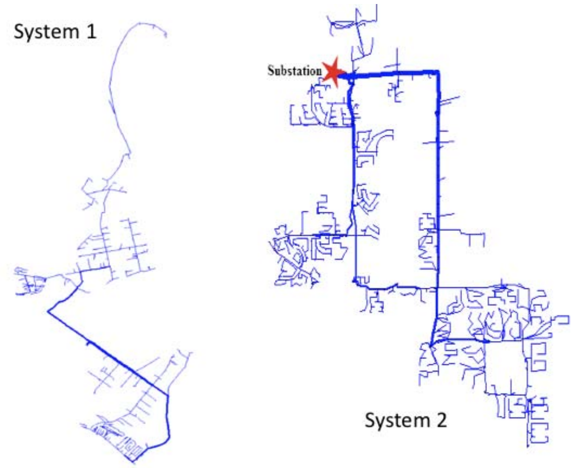

Fig. 6. Topologies of System 1 and System 2.

80% offset of the annual load consumption. For example, if a load consumes 10,000 kWh per year, an 80% load offset for PV means that during the simulated year, total PV generation output is 8,000 kWh. As a result, with the availability of normalized annual PV generation and load demand data, the kilowatt size of each PV system is determined by dividing the load consumption by the annual PV production per kilowatt PV rating given from the normalized PV profile, as:

$$P_{PV,i}^{DC} = \frac{\sum_{j=1}^{T} LoadN_i \cdot P_{load}^{i}}{\sum_{j=1}^{T} PVN_j} \tag{13}$$

where $P_{PV,i}^{DC}$ is the size of the PV panel in kW_{DC} to be added to the bus connecting to the *i-th* load, and P_{load}^{i} is the kW rating of the *i-th* load. $LoadN_j$ is the normalized load demand at time step j. PVN_j is the normalized PV generation at time step j. T is the total time points in the annual timescale.

It is worth to say the 80% offset of annual load consumption is used to determine each individual PV panel size if a customer load wants to install PV, but is not necessarily equal to the actual PV penetration in the feeder. This paper studies the 100% PV penetration scenario, i.e., the total kilowatt sizes of all existing PV systems equal to 100% of the peak loads. It is assumed that each load can have an aggregated PV system with the size determined using (13). PV system locations are randomly chosen from among all customer load buses to achieve 100% penetration.

All PV systems are assumed to have smart inverters. The DC-to-AC ratio is defined as the ratio between PV panel power rating (DC) and solar inverter power rating (AC). With PV panel size and DC-to-AC ratio confirmed, the corresponding solar inverter power rating can be determined. For economic reasons, solar developers often size inverters smaller than the PV panel nameplate kilowatts. The DC-to-AC ratio used in this paper is 1.25.

Both actual PV and load data measurements in year 2015 with 15-min resolution are obtained for two distribution systems, and these data are applied to PV and load models to capture the accurate PV fluctuation and load variability.

TABLE II
FIVE SCENARIOS STUDIED

	Description
Scenario I	No PV, No Voltage Optimization
Scenario II	100% PV, No Smart Inverter, No Voltage Optimization
Scenario III	100% PV, No Smart Inverter, Legacy Voltage Control
Scenario IV	100% PV, 100% Smart Inverter, Legacy Voltage Control and Autonomous VVC
Scenario V	100% PV, 100% Smart Inverter, Legacy Voltage Control and Aggregated Reactive Power Control

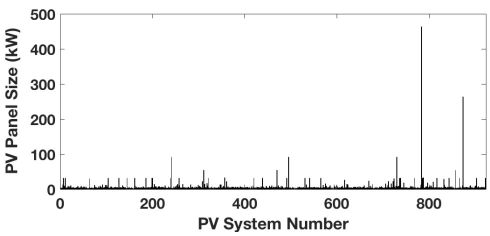

Fig. 7. The panel sizes of 922 PV systems in System-1.

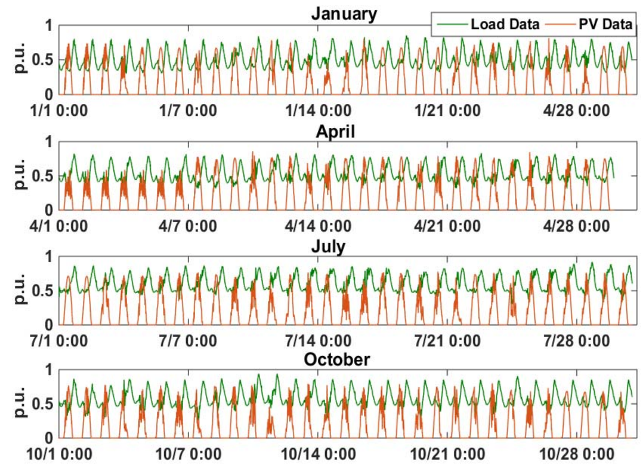

Fig. 8. Load and PV profiles for the selected days to study.

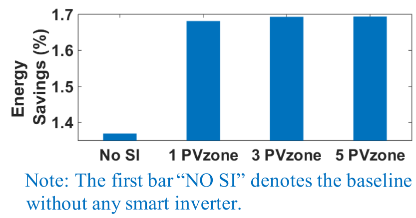

Fig. 9. Comparison of energy savings by choosing a different number of PV zones for Algorithm 2 with the aggregated smart inverter VVC.

B. Scenarios Studied

Four months—January, April, July, and October—are selected to cover four seasons, and then CVR energy savings for each month are obtained. A total of five scenarios are studied for all selected days, as shown in Table II. QSTS simulation with a 1-hour time step is conducted for the selected days to obtain the results for each scenario.

C. Simulation Study for System 1

1) Volt/VAR Curve Selection and PV Zones Definition: In total, 922 PV systems are added into the System 1 model, and Fig. 7 shows individual PV panel sizes. 71.9% of 922 PV systems are of sizes less than 5 kW, and 19.4% are between 5 kW and 10 kW. Two large PV systems, 463.3 kW and 263.5 kW, also exists in System-1. Total installed PV capacity is same as peak load power, i.e., 6.3 MW.

Fig. 8 shows the normalized load and PV profiles for the selected days. The three volt/VAR curve parameters used in the autonomous VVC (Scenario IV) for System 1 are given as: vvcCenter $=$ 0.96 p.u., vvcWidth $=$ 0.01 p.u. and vvcDeadbandWidth $=$ 0.002 p.u.

Before studying Scenario V, the number of PV zones should be decided. Three 1-day simulation scenarios are studied for solving the aggregated reactive power control problem by using different numbers of PV zones, including one zone, three zones, and five zones. Fig. 9 shows the results of energy savings obtained for three scenarios. Generally, with the participation of aggregated smart inverter control, the 1-day energy saving is increased significantly. The energy savings achieved by dividing the one, three, and five PV zones are 1.683%, 1.693%, and 1.694%, respectively. Because the scenario with three PV zones can obtain results that are quite similar to the scenario with five PV zones but it leads to less decision variables in (12), so three PV zones are finally chosen for studying Scenario V.

2) Simulation Results: Without adding PV systems or applying voltage optimization methods, i.e., under Scenario I, total load consumptions for January, April, July, and October are computed to be 2329.8 MWh, 2389.5 MWh, 2724.4 MWh, and 2587.7 MWh. Compared to Scenario I, Table III gives the percentage reduction on load consumptions obtained for all other scenarios. Scenario II results show that the integration of PV systems slightly reduces load consumption. Scenario III results show that the legacy voltage control can help reduce load consumption by approximately 1.7%–1.9%. Further, Scenario IV and Scenario V results show that the participation of smart inverters can help achieve additional reductions of 0.8%–0.9%, indicating that the additional use of smart inverter can help increase energy savings by around 50%. And a larger energy reduction occurs in Scenario IV, in which the autonomous VVC is implemented.

TABLE III
LOAD CONSUMPTION REDUCTIONS FOR DIFFERENT SCENARIOS

%	Scenario I	Scenario II	Scenario III	Scenario IV	Scenario V
January	--	0.109	1.881	2.770	2.647
April	--	0.029	1.879	2.737	2.645
July	--	0.059	1.707	2.661	2.507
October	--	0.043	1.724	2.686	2.535

Take January 1 as the example. Fig. 10 shows system voltage profiles obtained for five scenarios at 2 a.m. (no PV kW output) and 12 p.m. (high PV kW output), respectively. Without PV generation, the system voltage is generally decreasing as the distance from the substation increases. Instead, at 12 p.m. with high PV generation, the voltages of many buses are higher than substation voltages. As shown by the Scenario II results, the penetration of PV power will cause

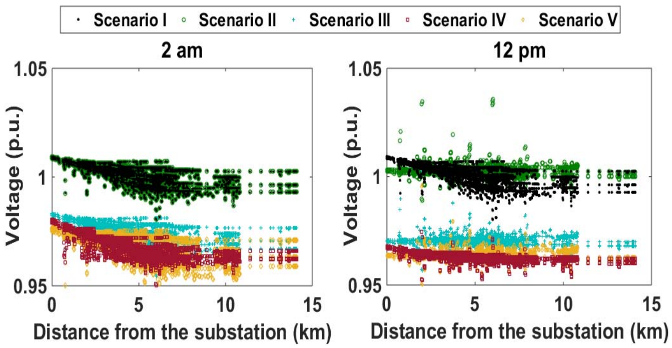

Fig. 10. System voltage profile for five scenarios on January 1 at 2 a.m. and 12 p.m., respectively.

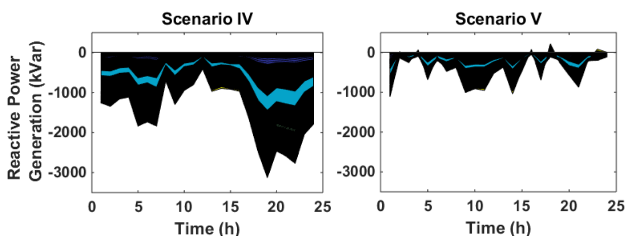

Fig. 11. Accumulated reactive power output from all smart PV inverters for scenario IV and scenario V, respectively.

the voltage to be up to 1.04 p.u., and this value will become bigger with larger PV penetration levels; however, with the implementation of CVR voltage optimization methods, system voltages obtained for scenarios III–V can be lowered significantly. And the system voltage profile in Scenario IV is tighter and lower than the one in Scenario V, which explains the results shown in Table III. The accumulated reactive power output from all PV systems in scenarios IV and V for each hour on January 1 are obtained, given in Fig. 11. It shows that much more reactive power is absorbed by smart inverters in Scenario IV.

Table IV shows the total energy losses obtained for five scenarios, in the unit of percentage to total energy consumption in each scenario respectively. The integration of PV systems can help reduce energy losses greatly, and the lowest energy losses occur in Scenario II, where PV systems are working under unity power factor and no voltage optimization is implemented. But the implementation of voltage optimization methods will cause some energy-loss increments. The sole use of legacy voltage control will cause very slight loss increments, and the addition of aggregated VVC will cause moderate loss increments, but the energy losses under these two scenarios are still less than Scenario I; however, the energy loss in Scenario IV becomes higher than Scenario I.

Table V shows the average substation power factors obtained for five scenarios, which are computed using the total active power and total reactive power generated from the substation. Compared to scenarios I–III, Scenario IV and Scenario V have lower substation power factors, which is because smart inverters absorb reactive power to lower voltages and thus more reactive power is needed from the substation.

TABLE IV
TOTAL ENERGY LOSSES FOR DIFFERENT SCENARIOS

%	Scenario I	Scenario II	Scenario III	Scenario IV	Scenario V
January	1.26	1.04	1.06	1.73	1.24
April	1.46	1.15	1.18	1.92	1.37
July	1.64	1.22	1.24	2.05	1.43
October	1.57	1.29	1.31	2.14	1.50

TABLE V
AVERAGE SUBSTATION POWER FACTORS FOR DIFFERENT SCENARIOS

	Scenario I	Scenario II	Scenario III	Scenario IV	Scenario V
January	0.975	0.952	0.955	0.675	0.869
April	0.974	0.946	0.951	0.661	0.858
July	0.974	0.952	0.956	0.685	0.881
October	0.974	0.953	0.958	0.686	0.884

TABLE VI
PV ACTIVE POWER CURTAILMENTS FOR TWO VOLT/VAR CONTROLS

%	Scenario IV	Scenario V
January	0.25	0
April	0.85	0
July	0.77	0
October	0.84	0

In addition, Table VI shows the PV active power curtailment for scenarios IV and V. No curtailment is needed for Scenario V, but PV active power is curtailed in Scenario IV to provide more reactive power.

Fig. 12 shows the accumulated number of LTC tap changes for five scenarios. In scenarios I and II, the LTC is controlled based on its default local settings. Without PV, 421 LTC tap changes are needed for a 4-month simulation, leading to 0.14 (421/2,952 time steps) tap change per time step on average. The integration of PV power causes the LTC tap changes to increase to 1,053, leading to 0.36 tap change per time step on average. The implementation of voltage optimization causes more LTC tap changes. The total number of tap changes for scenarios III, IV, and V are 2,292, 2,941 and 2,415, respectively, leading to 0.78, 1.0, and 0.82 tap change per time step on average.

D. Simulation Study for System-2

A total of 3,578 PV systems are added into System 2 to achieve 100% PV penetration for studying scenarios II–V, and three PV zones are used for Scenario V. The results of the load-consumption reductions, average system voltage, total energy losses, average substation power factor, and total number of operations of all legacy voltage regulating devices and PV curtailments for the 4-month simulation are provided in Table VII.

Load-consumption reduction for Scenario II is a negative value, indicating that the load consumption is slightly increased because of the integration of PV systems. Among scenarios III–V, both Scenario IV and Scenario V have more load-consumption reductions than Scenario III, but Scenario V gets more reduction than Scenario IV. Such results match with the results of average voltage, which shows that the aggregated approach causes the largest voltage reduction.

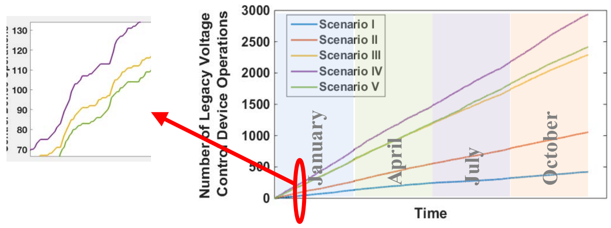

Fig. 12. Total operations of the legacy voltage control devices for different scenarios. Note: The zoomed-in figure is to illustrate that the increment is not simply linear.

TABLE VII
SIMULATION RESULTS OF FIVE SCENARIOS FOR SYSTEM II

	Scenario I	Scenario II	Scenario III	Scenario IV	Scenario V
Load Consumption Reduction	--	-0.142%	3.62%	3.95%	4.09%
Average Voltage	1.023 pu	1.023 pu	0.972 pu	0.970 pu	0.965 pu
Energy Loss (%)	0.94	0.75	0.70	0.86	0.59
Sub Power Factor[1]	-0.975	-0.808	-0.994	0.994	0.984
Legacy Volt. Ctrl. Device Operations	134	504	3,822	4,822	7,800
PV Curtailment	0	0	0	0.001%	0.009%

[1] A negative value means the power factor is lagging, whereas a positive value means the power factor is leading.

Compared to the Scenario II energy loss, both Scenario III and Scenario V can reduce energy losses, but Scenario IV causes energy loss to increase. Also, Table VII shows that the total operations of one LTC and seven capacitor banks are 134 and 504, respectively, for Scenario I and Scenario II, i.e., 0.0056 (134/2,952 time steps/8 devices) and 0.021 operation per time step for each device on average. The implementation of the voltage optimization approaches will cause more operations significantly, and the average operation for Scenario V is 0.33. Besides, both Scenario IV and Scenario V cause slight PV active power curtailment.

E. Result Discussion

The simulation studies on both System 1 and System 2 have demonstrated the effectiveness of the proposed CVR voltage optimization methods on saving energy consumption without violating system operational limits. The trends of the results are generally consistent between these two systems, but some mismatches exist because of the different system characteristics, including:

(1) Legacy voltage control achieves approximately 1.8% energy savings for System 1 but 3.62% for System 2. This mismatch can result from two reasons: (a) System 1 and System 2 respectively operate on 12 kV and 21 kV voltage classes, and the higher voltage likely contributes to higher energy savings because typically such feeders have heavy loads and more control devices. This phenomenon matches the conclusion summarized in [16]. (b) System 2 has seven capacitor banks, whereas System 1 has none. The presence of the capacitors can help achieve higher energy savings because they are used to flatten the voltage profile. With a flatter voltage, the LTC can tap lower, and thus more energy savings can be achieved.

(2) The participation of smart inverters can help achieve an additional 0.8%–0.9% energy savings for System 1 but approximately 0.3% for System 2. This mismatch also likely results from the existence of seven capacitors in System 2 because the effect of smart inverter cannot be demonstrated significantly if the voltage profile has already been low enough by using legacy voltage control.

(3) The autonomous smart inverter control achieves more energy savings than the aggregated control for System 1, but it is the opposite for System 2. Because System 1 does not have any capacitor, the coordinate operation of one LTC and three zonal smart inverter settings does not outperform the autonomous VVC in which each smart inverter adjusts its reactive power output based on its local voltage.

In addition to obtaining energy savings, this paper has analyzed the impact of CVR voltage optimization on energy loss, power demand from the substation, legacy voltage control device operation, and PV active power curtailment; and the simulation results of both System 1 and System 2 have shown that the CVR voltage optimization can bring negative effects on these terms. This is because the proposed CVR study considers energy consumption reduction as the only objective. Instead, if the metrics consisting of energy loss, reactive power demand from the substation, and control device operation are included into the optimization problem, it is expected that CVR energy savings will increase and power quality will improve simultaneously. We will address this problem by developing a multi-objective optimization problem in a separate paper in the future.

V. CONCLUSION

This paper studies the coordinated use of smart inverters with legacy voltage regulating devices to help increase CVR energy savings in distribution systems with distributed PV, and proposes two CVR voltage optimization models to address both autonomous volt/VAR control and aggregated reactive power control for smart inverters. The contribution of this paper includes:

(1) A repeatable approach based on detailed modeling and QSTS simulation is developed to evaluate and analyze the impact of smart PV inverters on CVR energy saving. DERs are recently allowed to participate into voltage regulation. Due to the lack of transparent studies and approaches, although both DER vendors and utilities are seeking for the opportunity of using DERs to improve distribution system operation, it is still unclear how to appropriately use DERs for distribution voltage regulation and how to effectively assess the impact of DERs after participating voltage regulation. The study presented in this paper can address such limitation.

(2) Two voltage optimization approaches are developed to incorporate autonomous VVC and aggregated reactive power control of smart inverters into the legacy voltage control that operates substation LTCs and capacitor banks. The proposed two voltage optimization approaches first address two types of smart inverter functions, and also provides the pilot method of complementing the traditional CVR scheme using smart inverter functions to maximize CVR benefit. The effectiveness

of the proposed approaches has been proved by the simulation studies on two real utility distribution systems.

In addition, the authors would like to address that this paper is not aiming at competing autonomous VVC and aggregated reactive power control based CVR approaches, or presenting a sophisticated voltage optimization approach to provide globally optimal solutions, but targets at accurately modeling and evaluating the impact of smart inverters on CVR energy saving and demonstrating the methods to achieve CVR benefit by using different smart inverter functions to supplement traditional approaches. The study presented in this paper can help utilities, vendors and researchers better understand the impact of smart inverters on CVR and power quality, which is still absent in existing studies but indispensable for developing effective and convex optimization models to coordinate smart inverters with traditional approaches to achieve maximum benefit.

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